

Trustworthy AI

- Week 5 Lecture 2
- Fairness Audit Lab
- Auditing and Mitigating Bias in AI Systems

Learning Objectives

- Train a simple classification model
- Measure fairness across groups
- Identify potential bias
- Apply a mitigation technique
- Compare before-and-after fairness results

Lab Scenario

- Hiring AI Audit
- Train a prediction model
- Measure fairness
- Detect bias
- Improve fairness
- Evaluate trade-offs

Famous US Dataset

- UCI Adult Income Dataset
- Predict Income > \$50K
- Sensitive attributes: Sex, Race, Age
- UCI:
<https://archive.ics.uci.edu/ml/datasets/adult>
- Kaggle:
<https://www.kaggle.com/datasets/uciml/adult-census-income>

Lab Workflow

- Dataset → Train Model → Measure Fairness → Identify Bias → Apply Mitigation → Re-evaluate Results

Step 1

- Train a Baseline Model
- Choose Any: Logistic Regression, Random Forest, XGBoost
- Record Accuracy, Precision, Recall

Step 2

- Measure Fairness
- Compare Male vs Female, Race, Age
- Metrics: Demographic Parity, Equal Opportunity

Step 3

- Identify Sources of Bias
- Historical Bias
- Sampling Bias
- Measurement Bias
- Model Bias

Step 4

- Apply a Mitigation Technique
- Suggestion: Reweighing OR Resampling

Step 5

- Re-Train and Re-Evaluate
- Compare Accuracy, Demographic Parity, Equal Opportunity

Reflection Questions

- Which fairness metric did you use?
- Was the model fair?
- Did mitigation help?
- Did accuracy change?

Suggested Libraries

- pandas, numpy, scikit-learn, matplotlib, shap (optional library)
- Videos: refer previously shared videos, else google them

Suggested Resources

- [Fairlearn.org](https://fairlearn.org)
- AIF360 documentation
- UCI Adult Dataset

Recommended YouTube Videos

- Fairness in ML:
<https://www.youtube.com/watch?v=NIlwexiljA>
[SA](#) (nice summary of what I taught in class)
- IBM Fairness in AI:
<https://www.youtube.com/watch?v=c4mRsoD>
[X150](#) (good case study to watch on how to fix bias)
- SHAP Tutorial from Week 4 (Optional)

Deliverables

- Submit notebook and 1–2 page reflection, all in PPTX
- Deadline: Next Tuesday 1159 PM as usual